

Answer key Exercise RSC-7: Air-Cooled Condenser

Performance

Source: Solution and Answer Guide, Silberstein/Obrzut, The Complete HVACR Manual 2E, Section 3: Refrigeration System Components (RSC), and the RSC-7 lab sheet.

Context

Objectives: Upon completion of this exercise, you should be able to perform basic evaluation procedures on an air-cooled condenser.

Introduction: You will use a thermometer and a gauge manifold to calculate subcooling and the condenser split (CS) of the system. A typical split is about 30° but can vary depending on the type of unit and its efficiency.

Takeaway: When evaluating a system component, such as the condenser, it is important for technicians to gather as much information about its operating conditions as possible. Subcooling and condenser split are two important calculations done when executing an evaluation of an air-cooled condenser.

Exercise RSC-7

RSC-7 Step 6, recorded data

Record the following: type of box (low, medium, or high temperature); type of condenser (natural or mechanical draft); suction pressure (psig); head pressure (psig); condenser saturation temperature (CST) (converted from head pressure) (°F); condenser entering air temperature (°F); condenser split (CS) (condenser saturation temperature – condenser entering air temperature) (°F); liquid line temperature (°F); condenser subcooling (condenser saturation temperature – liquid line temperature) (°F).

Model answer: The measured values depend on the system and have no fixed answer. Check that the calculations follow the lab sheet: condenser split = condenser saturation temperature – condenser entering air temperature; condenser subcooling = condenser saturation temperature – liquid line temperature. Type of condenser is natural draft (no fan) or mechanical draft (with a fan). *Not in the official answer key. The formulas come from step 6 of the lab sheet, the condenser types from model answer 6.*

RSC-7 Step 7

If the condenser split is more than 30°F, check for possible causes and note them here.

Model answer: Possible causes include a dirty condenser and air recirculating through the condenser, because recirculated air is warmer and causes high head pressure.

Not in the official answer key. Composed only from model answers 5 and 7.

RSC-7 Summary statement

Describe how condenser split and subcooling can be used to evaluate the operation of a condenser.

Model answer: Condenser split is the condenser saturation temperature minus the condenser entering air temperature. A typical split is about 30°F, but it can vary depending on the type of unit and its efficiency; a split above 30°F points to a problem such as a dirty condenser or recirculated air. Subcooling is the condenser saturation temperature minus the liquid line temperature and shows how far the liquid has been cooled below the condensing temperature, which is one of the three functions of the condenser. Together, the two calculations give information about the operating conditions of the condenser.

Not in the official answer key. Composed only from the Introduction, step 6 and the Takeaway on the lab sheet and model answers 1, 5 and 7.

Question RSC-7.1

Name the three functions that a condenser performs.

Model answer: Desuperheats, condenses, and subcools the refrigerant.

Question RSC-7.2

What is considered the highest temperature that the discharge gas should reach upon leaving the compressor?

Model answer: Approximately 210°F.

Question RSC-7.3

What may be used on an air-cooled condenser with a fan to prevent the head pressure from becoming too low in mild weather?

Model answer: A low-ambient control to cycle the fan.

Question RSC-7.4

What effects would low head pressure have on a system with an air-cooled condenser?

Model answer: The evaporator is starved because the pressure drop through the expansion device is too small for the correct flow.

Question RSC-7.5

Why must an air-cooled condenser be located where the air will not recirculate through it?

Model answer: Recirculated air causes high head pressure because the air is warmer.

Question RSC-7.6

What is the difference between a natural draft and a mechanical draft condenser?

Model answer: A natural draft condenser has no fan, it gives up heat to the surrounding area, a mechanical draft condenser has a fan that either pulls or forces air through the condenser to ensure heat transfer.

Question RSC-7.7

What are the symptoms of a dirty condenser?

Model answer: Typical: High suction and discharge pressures, low superheat and subcooling, and high compressor amperage.

Question RSC-7.8

What is the purpose of the fins on the condenser?

Model answer: To increase the surface area of the tubes to improve condenser heat transfer.

Question RSC-7.9

What materials are typical refrigeration condensers made of?

Model answer: Copper, aluminum, and steel.